DAY 24

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Group:-C2

Final Method and Validation Results

Dataset V1

- Total paired images: 312
- Train: 218
- Validation: 47
- Test: 47
- Image size: 256 × 256 RGB

Final Method: CNN

- Supervised paired image-to-image enhancement
- Trained using Dataset V1
- Learns mapping from underwater input → enhanced target
- Metrics: PSNR, SSIM, Edge F1

Classical Reference: Refined CLAHE-3

Method	PSNR	SSIM	Edge F1
Refined CLAHE-3	10.8626	0.4217	0.0656
CNN	14.2688	0.5517	0.0377

Final Reproducible Workflow

Raw Underwater Images
↓
Quality Check
↓
Duplicate Check
↓
Input–Target Pairing
↓
Dataset V1
312 Paired Images
↓
Frozen Split
218 Train | 47 Val | 47 Test
↓
Preprocessing
256 × 256 RGB
Normalization 0–1
↓
CNN Training
↓
CNN Inference
↓
Enhanced Output
↓
PSNR + SSIM + Edge F1
↓
Final Comparison
CNN vs Refined CLAHE-3

Reproducibility

- Fixed Dataset V1
- Fixed train/validation/test split
- Same evaluation metrics
- Test data kept separate
- Final configuration frozen



Final Technical Review and Next Corrections

Completed

- Dataset inspection
- Dataset V1 preparation
- Frozen train/validation/test split
- Classical baseline evaluation
- CNN training
- CLAHE refinement
- Robustness / ablation comparison
- Final candidate selection
- Pipeline freeze

Questions for Mentor

1. Sir is the final CNN selection acceptable?
2. Are PSNR, SSIM and Edge F1 sufficient for final reporting?
3. We Should additional visual/error analysis be included?
4. Should inference time and model size be reported?
5. Is there any corrections required before final report?

THANK YOU